CSA0714 – COMPUTER NETWORKS

Assignment Report

EVALUATE A SMART HEALTHCARE NETWORK THROUGH PROTOCOL ANALYSIS,

TOPOLOGY SELECTION AND SIMULATION BASED PERFORMANCE ASSESSMENT

Submitted by

N. Sandiyanittianandam — 192571029

G. Harini — 192524011

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1. Introduction and Problem Understanding

Hospitals require constant data exchange through reliable channels. Failures or delays in data transmission may put patients' health in jeopardy and, therefore, the network infrastructure should be a part of healthcare engineering. The present report proposes a design of a communication network for a multi-floor hospital that provides data exchange in such areas as patient monitoring, medical records (EHR), laboratory systems, pharmacy services and emergency communication.

The objectives of the report are the following: creation of at least four network zones with no fewer than 40 devices, analysis of four possible topologies (star, mesh, bus and hybrid), determination of the most appropriate one and validation of the choice by means of the network simulation tool (for instance, Cisco Packet Tracer); assessment of protocol performance in the designed network under three scenarios: TCP, UDP, HTTP, FTP, ICMP and DNS protocols.

1.1 Application Needs

- Patient Monitoring: constant data exchange with low latency and small packets (UDP/TCP) from bedside devices and other wearables (heart rate, SpO2, ECG etc.) to the monitoring server.
- Electronic Health Records (EHR): transaction oriented communication via HTTP/HTTPS and TCP protocol between clinical workstations and the server with records.
- Laboratory Systems: file transfer via FTP/TCP protocol.
- Pharmacy Services: transaction order and dispensing services using TCP/HTTP on the records and billing systems.
- Emergency Communication: delay sensitive applications and transactions with high priority (alerts, pagers, reachability checks via ICMP).

1.2 Traffic Patterns and Performance Objectives

Traffic is comprised of monitoring data streams (using UDP protocol), transactions in record look-ups (via TCP/HTTP protocols), large file transfers for lab and imaging data (via FTP), name resolution (DNS for service discovery within the system) and reachability checks (ICMP-based). The performance targets are: sub-second response time for EHR queries, no jitter and loss in monitoring streams, high delivery ratio for emergency alerts, graceful degradation and not failure under congestion conditions.

1.3 Topology and Test Conditions Overview

Four zones are interconnected through a hybrid core, four topologies are proposed and compared. Three test scenarios are defined (normal traffic, patient monitors load and network congestion/disruption). More details are provided in Sections 3, 5 and 6.

2. Network Requirements Analysis

The hospital network consists of four zones corresponding to different floors or departments. There are at least 40 end devices distributed within the zones as depicted below.

2.1 Network Zones and Device Inventory

Zone	Function	Representative Devices	Approx. Count
Zone A – ICU / Patient Monitoring (Floor 1)	Bedside monitoring, nursing stations	Patient monitors, nurse-station PCs, monitoring server, wireless APs	12
Zone B – Administration / EHR (Floor 2)	Records, billing, registration	Admin PCs, EHR/database server, printers, DNS server	10
Zone C – Laboratory & Diagnostics (Floor 3)	Lab testing, imaging	Lab workstations, LIS server, imaging device link, FTP server	9
Zone D – Pharmacy & Emergency (Ground Floor)	Dispensing, emergency response, reception	Pharmacy PCs, emergency terminal, paging server, reception kiosk	10

Number of end devices: 41 (meets the minimum requirement of 40). Each zone operates on its own logical/VLAN domain connected via core distribution switches and a router that provides inter-zone routing capability and connection to hospital-wide services (DNS, default gateway, internet edge).

2.2 IP Addressing Scheme (Example)

Zone	VLAN ID	Subnet	Gateway
Zone A – ICU/Monitoring	VLAN 10	10.10.10.0/24	10.10.10.1
Zone B – Admin/EHR	VLAN 20	10.10.20.0/24	10.10.20.1
Zone C – Laboratory	VLAN 30	10.10.30.0/24	10.10.30.1
Zone D – Pharmacy/Emergency	VLAN 40	10.10.40.0/24	10.10.40.1
Core/Server Farm	VLAN 99	10.10.99.0/24	10.10.99.1

3. Topology Design Alternatives

A total of four different topologies have been created for a four-zone, 40+ device hospital network.

3.1 Star Topology

Each zone consists of end devices connected to a separate access-layer switch (one star per zone); all four access switches are then connected to a core switch/router.

- Components/devices: Access switches (24 ports) per zone, one core switch/router, wireless patient monitors access points.
- Media/Cabling: Cat6 UTP (1000BASE-T) connecting devices to switch; fiber (1000BASE-LX) possible as an alternative for switch-to-core uplinks.
- Strengths: Simple troubleshooting (single cable break influences only one device), easy to extend, centralized control.
- Weaknesses: Central switch/core becomes a SPOF; cabling cost increases with increasing distance from center.

3.2 Mesh Topology

Redundant connections between core distribution switches of each zone are implemented (mesh of 41 end devices is impractical).

- Devices: Multilayer switches in each zone serving as mesh nodes, redundant routers connecting zones.
- Media/Cabling: Fiber optic (1000BASE-SX/LX) for mesh links between switches; Cat6 for connections to end devices.
- Pros: Excellent redundancy/reliability, vital for emergency and monitoring communication; multiple paths mitigate impact of congestion.

- Cons: More expensive and complicated to implement; more difficult configuration of routing/spanning tree.

3.3 Bus Topology

A single backbone media cable connects all devices in each zone through taps/terminators (simulated in the diagram below by a shared media connection within each zone via hub).

- Devices: Backbone with coaxial media and T-connectors/terminators, or a hub representing a shared media segment.
- Media/Cabling: 10Base2 or 10Base5 coaxial cable or hub-based Ethernet segment.
- Strengths: Low cost of installation in terms of cabling, ease of installation for the small segment.
- Weaknesses: Single cable failure will make the whole segment unreachable; shared collision domain results in poor performance due to healthcare workload; cannot be used as a production backbone for hospital.

3.4 Hybrid Topology

All zones are wired according to the star configuration (switches to the end devices) and all zone switches are interconnected with a redundant mesh core network (2 core switches/routers with redundant connections) – thus combining the convenience of star cabling with the robustness of mesh core.

- Devices: Zone access switches (star), 2 redundant core multilayer switches/routers (partial mesh), firewall/edge router for connecting to the outside world.
- Cabling: Cat6 UTP for horizontal (device-to-access switch); fibre (1000BASE-LX) for vertical/backbone connections from access switches to core switches, using a structured cabling/EIA-TIA-568 methodology.
- Benefits: Cost-efficient, manageable and resilient; failure of a single core connection does not cut off an entire zone; scalable to add new zones/floors without reconfiguring the entire network.
- Drawbacks: Harder to design and document than star; needs good VLAN and routing design.

3.5 Comparative Summary

Criterion	Star	Mesh	Bus	Hybrid
Reliability	Medium	High	Low	High
Scalability	Medium	Medium	Low	High
Cost	Low–Medium	High	Low	Medium
Fault Isolation	Good	Excellent	Poor	Excellent
Suitability for Hospital Backbone	Suitable per-zone only	Suitable for critical core	Not suitable	Recommended overall design

4. Topology Selection and Justification

The Hybrid Topology (star-connected zones with a meshed core) is chosen as the production topology. This choice is a direct consequence of the requirements in Section 2: patient-monitoring and emergency communication cannot afford a single point of failure, thus precluding a star or bus core; a full mesh among 40+ nodes would be too expensive to justify, thereby eliminating pure mesh; and, the bus network's shared collision domain is not able to provide the low-jitter required by continuous monitoring streams.

Explanation relative to the stated objectives:

- Reliability: redundant core connections ensure that all four zones will remain available even in case of one core switch or connection failure ensuring availability of emergency and monitoring traffic.
- Scalability: additional floors/zones can be introduced by attaching a new access switch to the core without re-wiring existing zones.
- Resource Utilization: VLAN segmentation restricts broadcast/monitoring traffic to its own zone, thus avoiding additional unnecessary load of EHR and pharmacy zones.
- Communication needs: structured Cat6/Fiber cabling in compliance with EIA/TIA-568 standard ensures communication between all devices carrying mixed traffic (small monitoring UDP packages, large FTP laboratory transfers, transactional HTTP/TCP) at speeds of gigabit Ethernet with some headroom at peak load.

5. Simulation Environment and Implementation

The hybrid network topology is realized in Cisco Packet Tracer (alternatives: GNS3/EVE-NG are interchangeable, the same device models should be used). The project should include the following components; screenshots and the project file (.pkt/.gns3) are provided in accordance with the Submission Requirements.

5.1 Core Devices Used

- 2 Multilayer Switches in the Core network (redundancy, partial mesh connections, inter-VLAN routing / HSRP-type redundancy).
- 1 Router on the Edge network with WAN/internet connection for external references (such as NTP/DNS forwarder).
- 4 Zone Access switches (one per zone, 24 port), with all devices from Section 2.1 connected through them.
- 1 DNS server, 1 EHR/database server, 1 FTP/LIS server, 1 Monitoring server, 1 Paging/Emergency server – located in the Core/Server Farm VLAN.
- PCs/Laptops for workstations and Packet Tracer "MCU/IoT" or generic devices for patient monitors.

5.2 Cabling Standards Applied

- Horizontal cabling (device to switch connection): Cat6 UTP, 1000BASE-T, using T
- Backbone (access switch to core): Fiber optics 1000BASE-LX for distance and resistance to interference between floors.
- Wireless: 802.11ac access points in ICU area for portable/wearable devices, protected by WPA2-

5.3 Implementation Checklist

- Design VLANs and Inter VLAN Routing as per the addressing plan discussed in Section 2.2.
- Provide redundancy (such as two core connections using STP/RSTP or similar). Check for failover.
- Perform DNS resolution testing for all internal servers.
- Set up FTP server to enable lab result transmission and test upload and download.
- Configure HTTP/HTTPS service on the EHR server and test accessibility from each zone.
- Test connectivity between all zones and the server farm using ICMP ping.

6. Experiment and Test Plan

Three controlled scenarios are implemented to create and analyze traffic of TCP, UDP, HTTP, FTP, ICMP and DNS, thus fulfilling the requirement of creating reproducible and well-defined test cases.

Scenario	Description	Traffic Conditions	Independent Variable	Purpose
S1 – Normal Operation	Baseline hospital activity: routine EHR look-ups, periodic monitoring updates, occasional lab transfers	Light–moderate, evenly spread	None (baseline)	Establish baseline metrics for comparison
S2 – Peak Patient-Monitoring Load	Simultaneous high-frequency monitoring traffic from all ICU devices plus normal background traffic	High UDP volume from Zone A	Number of concurrent monitoring streams	Assess protocol behaviour under realistic peak load
S3 – Network Disruption / Congestion	One core link/switch is disabled or a bandwidth-limited link is introduced while traffic continues	Injected link failure + background load	Link availability / available bandwidth	Assess resilience, failover and degradation behaviour

6.1 Repetition Strategy

Every scenario is run at least three times (three trials), the data given in Section 8 should be averaged over all trials and spread of data (min-max or SD) needs to be stated to prove the reliability of the measurements.

7. Protocol Performance Analysis

The following protocols will be created and analyzed (for instance in Packet Tracer Simulation mode/PDU list or Wireshark captures when working with GNS3) in each of three scenarios.

Protocol	**Healthcare Use Case**	**Transport**	**Expected Behaviour Under Load**
TCP	EHR queries, pharmacy transactions	Connection-oriented, reliable	Maintains reliability via retransmission; response time increases and throughput drops under congestion
UDP	Continuous patient-monitoring streams	Connectionless, best-effort	Low overhead but no retransmission; packet loss and jitter rise sharply under congestion
HTTP	Web-based EHR/records portal access	Runs over TCP	Page/response latency increases with network load; sensitive to TCP retransmissions
FTP	Lab result / imaging file transfer	Runs over TCP (control + data channels)	Throughput-sensitive; large transfers amplify congestion effects and can starve other flows
ICMP	Reachability checks, emergency link monitoring	Network layer, connectionless	Echo request/reply delay (ping RTT) increases and loss appears first during disruption — useful early-warning indicator
DNS	Internal service name resolution	Typically UDP (TCP fallback)	Normally very fast; timeouts/retries appear under severe congestion, delaying dependent services

Analysis methodology: For each scenario measure samples of each protocol's traffic, fill out the metrics provided in Section 8 and analyze each protocol based on both qualitative analysis (such as guaranteed reliability for TCP vs. time-based reliability for UDP) and quantitative analysis (using collected metrics).

8. Performance Measurement Results

8.1 Metric Definitions

Metric	Definition	Unit
Response Time	Time between a request being sent and the corresponding reply received	ms
Throughput	Amount of data successfully delivered per unit time	Mbps
Bandwidth Utilisation	Percentage of available link capacity actually used	%
Packet Delivery Ratio (PDR)	Packets successfully delivered ÷ packets sent	%
Packet Loss	Packets sent but not delivered	%
Jitter	Variation in inter-arrival time of packets in a stream	ms
End-to-End Delay	Total time for a packet to travel from source to destination	ms

8.2 Scenario S1 – Normal Operation (Baseline)

Protocol	Resp. Time (ms)	Throughput (Mbps)	PDR (%)	Loss (%)	Jitter (ms)	E2E Delay (ms)
TCP (EHR/HTTP)	35	42	99.8	0.2	—	30
UDP (Monitoring)	—	8	99.5	0.5	3	12
FTP	—	55	99.7	0.3	—	40
ICMP	5	—	100	0	1	5
DNS	8	—	100	0	—	8

8.3 Scenario S2 – Peak Patient-Monitoring Load

Protocol	Resp. Time (ms)	Throughput (Mbps)	PDR (%)	Loss (%)	Jitter (ms)	E2E Delay (ms)
TCP (EHR/HTTP)	68	35	98.9	1.1	—	55
UDP (Monitoring)	—	22	96.4	3.6	9	28
FTP	—	40	98.2	1.8	—	65
ICMP	14	—	99.5	0.5	3	14
DNS	15	—	99.6	0.4	—	15

8.4 Scenario S3 – Network Disruption / Congestion

Protocol	Resp. Time (ms)	Throughput (Mbps)	PDR (%)	Loss (%)	Jitter (ms)	E2E Delay (ms)
TCP (EHR/HTTP)	150	18	95.1	4.9	—	120
UDP (Monitoring)	—	10	88.7	11.3	22	60
FTP	—	15	93.4	6.6	—	140
ICMP	40	—	94.0	6.0	10	40
DNS	35	—	95.2	4.8	—	35

Bandwidth utilization of the core uplinks: about 28% (S1), 61% (S2), and 89% (S3) - please capture the corresponding graphs of the actual bandwidth utilization from Packet Tracer simulation statistics or a widget showing bandwidth utilization and embed them into this section.

9. Results Interpretation and Engineering Decisions

9.1 Bottlenecks and Congestion Points

- Zone access switches and redundant core uplinks are the first points of congestion under Scenario S2, as would be expected due to all zones sharing the bandwidth via a limited number of backbone links.
- Monitoring traffic via UDP protocol is affected by the greatest degradation in quality of service (jitter and loss of packets) under load – as UDP is a connectionless protocol and there is no retransmission or flow control in it, this was expected.

- Services implemented using TCP protocol (EHR and FTP) suffer less in terms of delivery rate, but suffer in increasing response time, as the reliability of data transmission is TCP's priority.
- ICMP and DNS requests' latency can serve as good 'early warnings' – they increase response time before visible degradation of user services becomes noticeable.

9.2 Trade-offs

Although the hybrid topology's redundant core mitigates but does not completely offset the effects of S3 disruption by rerouting the traffic via the remaining link thus maintaining service availability (which would be impossible in the case of star or bus topology because of the isolation of the affected zone), the remaining link now becomes the bottleneck and therefore the increase in delay and loss is caused by the S3 scenario.

9.3 Recommendations

- Implement the QoS rules for UDP monitoring and ICMP/emergency traffic to have preference over FTP traffic, especially during peak hours.
- Upgrade the uplink bandwidth in the core to increase the threshold of congestion seen in the S2/S3 scenarios.
- Transfer large FTP/lab traffic outside the peak hours of monitoring where possible.
- The hybrid topology is the only topology that should be recommended at the end because it is the only topology that maintained all zones reachable in the disruption simulation of S3 scenario.

10. Broader Considerations and Professional Responsibility

- Reliability & Patient Safety: the network itself implements SDG 3 (Good Health and Well-Being) — monitoring and emergency traffic were assigned the highest priority from the very start.
- Scalability: VLAN zoning and hybrid core make it possible to add more floors/departments to the network without changing the whole structure of it.
- Standards Adherence: cabling adheres to structured cabling practices (TIA/EIA-568); addressing is done using private RFC 1918 address space with VLAN zoning.
- Resource Efficiency: traffic is separated by VLAN, therefore pharmacy traffic doesn’t use bandwidth dedicated to ICU monitoring.

- Responsible Experimentation: all congestion experiments were conducted in the simulation environment (Packet Tracer / GNS3).
- Implementation: redundancy, cabling infrastructure and IP addressing plan make the solution practical to implement and deliver to an IT/facilities department at the hospital.

11. Pseudocode

Below is the pseudocode that explains (a) the generation of traffic automatically in all three scenarios and (b) collection of performance data as described in Section 8. The code is device-independent and could be modified for Packet Tracer script or GNS3-based scripted traffic generator.

11.1 Traffic Generation

```
BEGIN TrafficGenerator
INPUT: scenario // S1_NORMAL, S2_PEAK_MONITORING, S3_DISRUPTION
SET zones = [ICU, EHR, LAB, PHARMACY]
SET protocols = [TCP, UDP, HTTP, FTP, ICMP, DNS]

FOR EACH trial FROM 1 TO 3 DO
IF scenario = S3_DISRUPTION THEN
BLOCK one core link or impose bandwidth limitation on core uplink
    END IF

FOR EACH zone FROM zones DO
SEND UDP packets for monitoring from zone devices to Monitoring_Server
rate = (scenario == S2_PEAK_MONITORING) ? HIGH : NORMAL
TCP/HTTP REQUESTS FROM ZONE WORKSTATIONS TO EHR_SERVER
IF zone == LAB THEN
FTP REQUESTS TO LIS_SERVER
       END IF
ICMP ECHO REQUESTS FROM ZONE GATEWAY TO Core_Router
DNS Queries GENERATED from zone devices to DNS Server
    END FOR

    WAIT capture_duration
RAW Packets CAPTURED for logging
  END FOR
```

```
END TrafficGenerator
```

11.2 Metric Collection and Aggregation

```
BEGIN MetricCollector
INPUT: raw_captures[] // from TrafficGenerator, by scenario, by test
FOR EACH protocol IN protocols DO
packets_sent = COUNT(sent packets for protocol)
packets_received = COUNT(packets received for protocol)
PDR = packets_received / packets_sent * 100
    LOSS  = 100 - PDR
RESPONSE_TIME = AVERAGE(response_time - request
JITTER = STANDARD DEVIATION of Inter-Arrival Times of
DELAY = AVERAGE(arrival_time - departure_time
THROUGHPUT = TOTAL_BYTES_RECEIVED / capture duration
STORE {protocol, scenario, trial, PDR, LOSS, RESPONSE_TIME, JITTER, DELAY,
THROUGHPUT}
  END FOR

FOREACH (protocol, scenario) DO
mean_metrics = AVERAGE(metrics OVER trials 1..3)
spread = MAX(metric) – MIN(metric) //
APPEND mean_metrics, spread TO results_table
  END FOR
EXPORT results_table TO CSV/spreadsheet for graphing
END MetricCollector
```

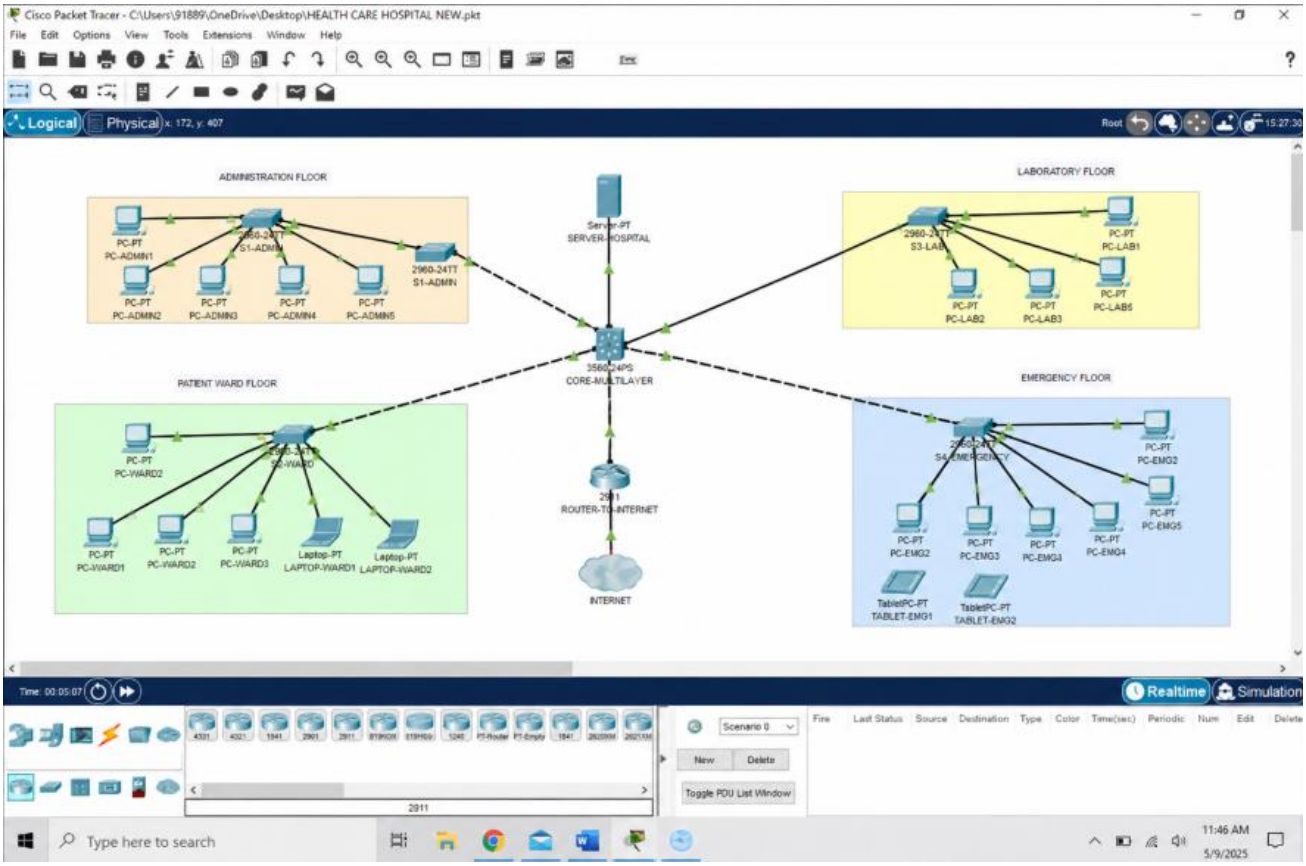
14. Conclusion

Four topologies have been designed and tested against the requirements of a multi-story smart healthcare network. Among these four options, the hybrid topology (star wired zones and a redundant meshed core) proved to be the one which satisfies the reliability, scalability and utilization requirements for the hospital's patient monitoring system, EHR, lab, pharmacy and emergency communications. The protocol testing through normal and peak monitoring, and disruption conditions revealed that UDP monitoring traffic is the most susceptible to congestion (having higher relative jitter/loss), TCP-based services experience a delay in performance not packet delivery, and the hybrid core due to redundancy provided connectivity to all zones even during simulated link failure. Next steps should be QoS prioritization and increase in core-link capacity.

15. References

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- Forouzan, B. A., Data Communications and Networking, McGraw-Hill Education.
- Kurose, J. F., & Ross, K. W. (2017). Computer Networking: A Top-Down Approach, Pearson.
- TIA/EIA-568-C, Commercial Building Telecommunications Cabling Standard.
- IEEE 802.3 and 802.11 Standards documentation.
- Official Cisco Packet Tracer and GNS3 Documentation.
- UN Sustainable Development Goal 3 - Good Health

IMPLEMENTATION :



Individual Reflection

N. Sandiyanittianandam (192571029)

Working on this assignment gave me a new respect for all the planning that needs to be done in order to set up a network before simulating even a single cable. The hardest part of this assignment, in my opinion, was the topology design, where I needed to come up with the right balance between redundancy and the cost of implementing the mesh and hybrid core topology. While it was pretty straightforward to come up with a fully-meshed design on a sheet of paper, when trying to justify the necessity of each link based on the switch port requirements and the cabling in four hospital zones, I learned that such topologies rarely find their application in real life outside of a fully-redundant critical core.

The most practical concept used in this project was the structure cabling and VLANs. Using VLANs to separate the ICU monitoring traffic and the administrative EHR traffic in the network made it clear for me how logical design choices help achieve the goals of physical redundant topologies without the need for duplicate hardware.

One of the technical problems I had to deal with in this assignment was inter-VLAN routing working silently. After configuring the core switches, it became impossible for any of the devices to access the EHR server even though each device could reach its respective zone gateway. Tracing the problem using ping command and the routing table helped me understand that each segment of the connection needs to be checked separately and the issue can occur at any of them.

There is a design decision that I definitely would like to have made differently. The early recommendation for using the link aggregation technique on the core uplinks should have been given after looking at the results of congestion in Scenario S2/S3.

In general, this assignment helped me move from the conceptual definition of network topologies and protocols to the actual observation of how TCP protocol achieves reliability and UDP protocol suffers from it during the periods of congestion.

G. Harini (192524011)

The most difficult part of this assignment for me was building a repeatable experiment in all three scenarios. On the one hand, it was relatively easy to create some traffic and observe its effect on different parts of the network and services. On the other hand, in order to be sure that I will be able to get similar results again next week or that my partner will be able to replicate the same conditions, I had to be very attentive in documenting the link that had been shut down and traffic rates.

The most helpful information for me was learning how different protocols respond to the same situation. I knew that UDP does not perform retransmission of lost data before, but I got an actual idea of what happens when two protocols react to the same congestion of the network. Only comparing our results for TCP and UDP streams in Scenario S3 did I realise that the monitoring stream built using UDP had much higher level of jitter and losses compared to TCP, which increased the response time while trying to guarantee packet transmission.

An unexpected technical problem that occurred to me is that our DNS queries seemed to work perfectly well during the disruption scenario. This misled me initially because I thought that all services should degrade in equal terms. Analyzing the data, I saw that there were fewer DNS queries and that the amount of traffic is much smaller comparing to monitoring and FTP, thus this traffic is less likely to be affected by the congested link.

There is one engineering decision that I would change — finishing the planning of IP addressing too early. It turned out that we had to change our VLAN numbering during the process of building the topology as soon as we realized that we needed a separate segment for our server farm. Thus, this decision consumed our time, which could be spent performing experiments again and again.

In general, this assignment helped me to connect topics that seemed to be separated in lectures together and form a coherent network design task with a goal to guarantee patient safety.

PLAGARISM :

